

Futuring from an indigenous community stance: projecting temporal duality from the past into the future

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Figure 1: Portrayal of futuring sessions in Donkerbos

ABSTRACT

This paper presents the first instance and experience of futuring with a rural San community from the Kalahari desert in Donkerbos, Namibia. Over a series of sessions we explore divergent speculative design and design fiction methods to stimulate and invoke alternative green energy use cases. These alternatives are premised on the imagination of unorthodox green energy use, superseding interventionist energy use which is constantly propagated and mainstream. We showcase the application of speculative design and design fiction in challenging the dominant interventionist approach and singular temporal view, resulting in a dissentient dual temporality. As well as demonstrate its utility and inadequacies in transitioning an African rural indigenous community into the speculative, arguing for the appropriation and widening of futuring methods in an African context.

CCS CONCEPTS

• **Human-centered computing** → *Empirical studies in HCI*.

KEYWORDS

Futuring, Indigenous participants, Namibia, Temporality

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1 INTRODUCTION

Many rural African communities face extreme energy poverty and backlog through exclusion from national energy systems, with an estimated 600 million people excluded from stable and reliable energy access [33]. Consequently local communities compensate and utilise unsustainable, inappropriate and limited energy sources. Meanwhile, green energy projects and initiatives propagating green energy access have been established across the continent [5, 35]. However, most of these projects follow an established rural African development approach, based on declared deficits and needs, with solutions derived and deployed, mostly by external entities. It implies, that mainstream energy uses such as lighting, cooking, cooling, water pumping and gardening are championed [3, 43]. However, these use cases do not move the needle in terms of rural innovation. Rather, they epitomise solutions that are ‘obvious’ and conventional, perpetuating interventionism [1]. This approach is premised on solutionism and saviourism, with “solutions and saviours” coming from elsewhere to save Africa [1, 52]. Viera and Couldry [53] point

to the cunning phrasing and exaggeration of such solutions as the only viable and available panacea. However, such interventions operate on a kernel of slanted dependencies and disparities bordering neo-imperialism [16, 19, 34], with solutions reflective of an idealised Western sense of mobility and progress. Despite counter efforts (see e.g. Cunningham et al. [12], Okeke et al. [37]), there has been little emancipation and autonomy from the hierarchy and dominance of a Westcentric approach, especially in the digital technology and green energy space. This, consequently, results in the dominant reproduction and espousal of a European model of life, impeding the ascendance of genuine alternatives [31].

As researchers from the Global South, working with local communities, we argue and necessitate the emergence of *bona fide* alternatives. Thereby, we initiate futuring activities with a rural San community in Donkerbos, Namibia, as a way to stimulate and invoke alternative green energy use cases based on the principles of what Leitão [31] calls “agentic desires”. In this paper, we showcase the application of futuring processes, speculative design and design fiction, in challenging the dominant interventionist approach and singular temporal view, as well as its utility and inadequacies in transitioning an African rural indigenous community into the speculative.

2 THEORETICAL FRAMING

2.1 Futuring

As stated by Dunne and Raby in Design Noir [14], “All design is ideological, the design process is informed by values based on a specific world view, or way of seeing and understanding reality. Design can be described as falling into two very broad categories: affirmative design and critical design.” One of those approaches being sceptical towards the current trends is the ‘Futuring’ method. Defined by Hoffman et al. [25] as the active process of imagining the future, futuring was first proposed by Ossip Flechtheim in the mid 1940s as part of the future studies narrative [7]. In recent times, it has provided utility in different domains through the actualisation of methods such as speculative design and design fiction [20, 32]. These methods have found popularity in design and HCI as reliable and valuable tools for thinking and imagining the future. They have allowed industrial designers and researchers to explore the future, establishing future visions [24]. Thus they are providing capacity to determine what the future could look like, with an emphasis on transcending towards that future. This has not always been the case, given the possible constraints, limitations and scepticism towards futuring [28]. However, Balcom Raleigh and Heinonen [7] elaborate the utility and merit of participatory futuring in allowing participants to disengage from object limitations and daily constraints.

Dunne and Raby [15], credited for speculative design and design fiction, apply and advocate for futuring in a wide range of domains and demonstrations. Accordingly, Snow et al. [44] demonstrate the use of speculative design to engage and understand participants’ values regarding future energy use, positioning users centrally in the design of future energy technology. Their study indicates how speculative design can invoke awareness of complex and concealed issues, allowing extended participation. Whilst Chopra et al. [10], contributing to participatory speculative design, in their study with

grassroots food growing communities explore and negotiate sustainable collaborative futures. Through a series of workshops, they employ mapping, walking, gaming and making. Particular emphasis was placed on understanding and integrating local knowledge, situating speculation, gamified future building and making apprised by local narratives and values.

Further, speculative design methods such as *what if* have been used and explored, as demonstrated in Farias et al. [20]. Design fiction was used by Sturdee et al. [48] to build a fictional world premised on algorithm-enabled empathy as a central factor in future communication. They orient their work around a diegetic prototype to suspend disbelief about the world. It was built to encounter questions and skepticism about the plausibility of the world they present. The above studies, and others, contribute to the popularity and uptake of the methods, with continuous advocacy for their application in different domains, spaces and contexts [13, 27]. Yet, Soden et al. [45] in their literature review surveying futuring work and projects globally show the lack of futuring endeavours in Africa with projects reported only in relatively privileged settings in Europe, United States, Latin America and Asia. Meaning that futuring primarily remains in its comfort zone, “inattentive to its own biases or, at worst, elitist” [30].

2.2 Temporality of futures

In his book exploring the underlying fundamentals and complexities of the future, John Urry states that the future “remains a mystery”, arguing that it is rarely a simple extrapolation of the present [51]. Therefore, it is a white canvas of possibilities and temporalities [38]. Yet, it is often portrayed through a single view, as evidenced by the common application of the *Futures cone* [11, 23]. Despite providing a descriptive utterance of different types of futures, the cone, according to Kozubaev et al. [30] presents the future progressing from “a single point called ‘the present’ on the left, towards multiple futures on the right”. This implies “a shared present, which can obscure complexities of historical context as well as the diversity and situatedness of presents” [30]. Therefore, dominant conceptualizations of temporality needs to be challenged [40], allowing for due consideration of non-conformist and alternative understandings of temporalities.

Designers are paying attention to and analysing different temporal perceptions [40, 54]. Specifically, HCI’s temporal sense has begun to widen, suggesting “a plurality of perspectives and experiences of time” [42]. With this, HCI researchers are moving away from an objective and “absolute” stance of time to a more “relative” and context oriented perspective [41]. This is in line with the call by Pschetz and Bastian [39] to include and expand “alternative temporalities that are neglected by the dominant narratives”. Taylor et al. [49] demonstrate this in their work with an Australian Aboriginal community, highlighting the community’s distinct perspective and cultural nuance of time. Whereas, Friedman and Yoo [22] explore time from a multi-life dimension, prescribing for methods and designers to take into account multiple lifespans. While, Chen [9] and Odom et al. [36] in their work, aiming to demonstrate the diversity of temporal design and expand temporal considerations, examine and postulate temporality within a frame of “slowness”.

These cases, amongst others, demonstrate the gradual and imperative shift towards diverse, multiplex and meaningful concepts of temporalities substantiating the *Pluriverse* as denoted by Escobar [18]. Despite this, African temporalities are often disregarded and misinterpreted, giving a distorted view of its chronicle and people [2], affecting the design of alternative technologies and futures.

3 EXPLORING FUTURING WITH AN INDIGENOUS COMMUNITY IN THE KALAHARI

3.1 Introducing the community

Donkerbos is a remote resettlement village farm in the Omaheke Region, East Namibia. A close-knit of 200 San people from different San clans reside on the farm. The San, indigenous to southern Africa, are part of the Khoe group and speak the X'ao-'aen and Naro languages. They are among the oldest inhabitants in the world with a nomadic and hunter-gatherer lifestyle dating many millennia when their ancestors trekked the Kalahari Desert. However, with colonisation, apartheid and Namibia's independence, their freedom, autonomy, nomadic lifestyles and culture were interrupted with acres of land and wildlife taken away and fenced off, in addition to other oppressive laws; consequently leading to resettlement on farms, as is the case with Donkerbos. Resettlement farms have many restrictions, with limited land, wildlife and heritage. Thus the Donkerbos community has lost many of its cultural practices and identity, adopting a conventional lifestyle favouring modern ethos and amenities. However, there has been little inclusive or appropriate integration and development, with a majority of the San marginalised, mistreated and limited access to opportunities. Since 2018, a local university has been collaborating with the community on various initiatives and projects, among others exploring emerging technologies such as augmented and virtual reality in a rural and cultural context [6, 55], in addition to helping the community set up a campsite and living museum, and apply for grants. Despite these efforts, obstacles remain. As with many other African communities, the community faces energy constraints, leading to the initiation of energy access projects and initiatives. Yet most projects prioritise mainstream developmental and interventionist approaches, resulting in often unsustainable and non-innovative use cases. This was the case in Donkerbos, with a previous energy project providing solar energy systems to households. The systems were used finitely with limited and basic use such as lighting and cellphone charging. Thus, in an effort to establish green energy use alternatives and possibilities, we conducted futuring sessions as a step towards inspiring alternative, innovative and unorthodox rural energy use cases that are meaningful and productive.

3.2 Introducing sessions

Over a two-day period, we conducted four futuring sessions in Donkerbos (figure 1). The purpose of the sessions was to establish the community's perspective and understanding of the future, suspend the status quo and imagine alternative futures and technological possibilities. Each session had a different format and objective. The first session focused on identifying the cultural situatedness, temporality and meaning of future. The second session explored

the *what if* method as a means of suspending current constraints and transcending into imagination and speculation. The third session was a futuring walk with a focus on situated speculation and fictional scenarios. The fourth session was based on a diegetic exploration, with locally inspired digital games as prompts, showing plausibility of the alternative.

Before the sessions, there was an introductory meeting during which 14 community members, hereafter referred to as participants, autonomously chose to be part of the sessions. They comprised of old, middle-aged and young community members, with a female majority (9 women). They provided consent to capture data. Therefore, allowing us to capture video and still imagery using a digital camera and video recorder, in addition to notes taken during and after the sessions. Additional data was collected through several sketches, with participants, at times, preferring to graphically illustrate and present perspectives and concepts.

3.3 Session 1: Understanding the future

Inspired by Chopra et al. [10] future mapping, our primary objective was to formulate the community's conceptualization of temporality and perspectives regarding the future. We did this through a series of questions and queries about the meaning and their considerations about the future. Lasting 2 hours, we arranged the session as focus group discussion, with participants and researchers sitting circularly, allowing for collective discussions. We explained and emphasised the aim of the session, whilst simultaneously encouraging open and inclusive dialogue. With each question, participants shared and amalgamated their contributions, agreeing upon a common perspective and forming one concise response. Additional modes of expression were encouraged with sketches highlighting their viewpoints, concepts and aspirations at the forefront.

Responding to a question about what the future is, participants, misinterpreting the question, initially talked about what they want and need. However, after clarifying our interest in their temporality and worldview, they asserted that the future is "tomorrow, something that is coming and that you are going forward". Sensing reticence, we encouraged additional responses, resulting in emphasis that the future is linked to the past, because when thinking about the future, they think of something good that has happened in the past, and think about having it again in the future. At this point, showing elevated interest, participants became more engaged and involved. Daniel, a community member, relating the past to the future, asserted that "if there was something in the past that I did that brought success to me, I can apply it in the future so that it can bring something to me. It's just like planting something today, and then harvesting tomorrow". Thereafter, we explored the origin of ideas about the future by asking participants who and what influences and determines the future. In response, participants highlighted the central role elders play in providing guidance and reinforcement on how the future should be like, especially regarding behaviour, actions and character. However, the dynamics that inform the future and its possibilities from a community perspective were still unclear. Therefore, we asked what defines the future. Participants explained that there are two types of futures; a traditional and modern future. Denoting their attachment, they mentioned that "we have an emotional connection to the traditional future and at the same time a

desire for the modern future, yet lost in the present”. Elaborating their stance, showing great eagerness, participants drew sketches highlighting the central particulars of each future. The sketches illustrated the hunter-gatherer lifestyle premised on what their forefathers envisioned as ideal and desirable. Whereas the modern future showed contemporary utilities and livestock farming, which they envision in their desultory present, indicating the temporality and worldview of the future from the community’s perspective.

3.4 Session 2: What if?

Following the first session, we initiated a *what if* activity inspired by the *what if* method popularised by Dunne and Raby [15]. During this session, lasting half an hour, we asked participants *what if* questions in an attempt to expand thought and imagination beyond their current aspirations and perceptions of reality, anticipating the gradual imagination of hypothetical green energy use scenarios. The questions were based on the responses from the first session, with the question ‘if you had everything from the traditional and modern future what else would you want?’. An open and inclusive dialogue was emphasised, encouraging a wide range of responses, so that participants can generate unorthodox possibilities. To aid their deliberations, we asked additional prompt questions that would allow them to break the imagination barrier. We also provided examples of emerging and unorthodox technological artifacts that the community can consider when imagining future possibilities.

Initially, participants were unsure about the *what if* questions, showing dubiety. To them the answers to such questions were obvious and straightforward, with participants stating that “if you have everything you need you will not need anything more”. Further stating that “if you already have everything, you have no needs or idea of other things because you already have what you need. If you have everything, then you are fine”. They questioned the sense and appropriateness of the method, highlighting the flaws of the method, indicating that the questions did not make sense as they were formulated. The questions portray one as having everything and therefore in need of nothing, thus making imagining ‘what if this or that happens’ irrelevant. We reiterated the purpose of the method and questions, encouraging participants to come up with novel ideas and emerging possibilities, in addition to providing emerging technology examples. However, it was convoluted, with one participant asking “how will you think about something which you have not seen? If you have not seen something then you won’t have an idea of what to think. Also, if you get an idea there is no manufacturer that will be able to do it. Thus, you should be the first one to start with it”. Realising reluctance, we requested participants to state any arbitrary ideas that they may have. However, there were none forthcoming.

3.5 Session 3: Walking the future

Moving from the *what if* method, we explored spatial and situated green energy enabled possibilities, drawing inspiration from local elements through a one hour futuring walk (figure 2). Feltham et al. [21] indicate the benefits of walking as a mechanism in stimulating creativity and new ideas, thus providing a platform for imagination. Considering the community’s intrinsic relation to nature and

the environment, the walk was seemingly appropriate to invoke imagination and inventiveness. Participants devised a route for the walk, with the living museum as the central sight and starting point, in addition to other structures and sights. The intention was for participants to identify local ecological elements which can be used as a basis for future possibilities and technologies, allowing them to imagine alternative futures. Additionally, we also encouraged embodiment and manifestation of new ideas through illustration and simulation.



Figure 2: Session 3: Demonstration of San vulture flying styles (left), Future walk (right)

The walk was set along the community campsite, starting at the living museum. Walking as a group, we asked participants to identify sights and natural elements that present or enable unique green energy possibilities. We emphasised that the ideas should be unorthodox. At the first sight, the living museum, we asked them to imagine new ways that green energy can be used around the museum, possibly thinking of ways to enhance and elevate it. Sensing uncertainty and lack of input, we encouraged participants to share any thoughts and ideas that they had, even if they were crazy, as that was precisely the purpose of the exercise. However, there was hesitance, with participants not indicating any signs or sense of imagination. We suggested that they look at the trees or the sky for inspiration. Thereafter, a participant started looking at a nearby tree and suggested that colourful lights and digital screens be installed on the tree, with the lights denoting wild animals (such as a gemsbok, warthog and elephant) that used to eat the tree, and also show the use of the tree for medicine, roots, food, etc. Others were in agreement with the idea, stoutly emphasising the cultural and educational relevance of such a tree. Sensing elevated interest and imagination, we probed for additional ideas. However, there were no further contributions, with participants seemingly inarticulate. Therefore, we walked further into the bush, encouraging more ideas and asking participants to think and imagine new possibilities as we walked. We stopped, with little success, at several sights hoping to formulate new ideas as we gazed the mountains, trees and grassland. With ideas lacking, we prompted participants with examples, suggesting some type of robot based on their experience with technologies such as virtual reality. The suggestion, seemingly resonating with participants, resulted in two concepts, a San exhibition robot that would travel across different villages teaching and showcasing San culture, and a San flying vulture suit that gives community members the ability to fly. Embodying and manifesting the use of the flying vulture, participants demonstrated flying, running and lifting themselves up (as take off), spreading their arms in the air moving them up and down, similar to a vulture in flight. At this point participants were highly engaged, eagerly

and enthusiastically showing their own flying styles. They were immersed in imagination in a seemingly relaxed and carefree state, remarking that the session “allowed us to think of new ideas such as the San flying vulture”. Further stating that “we did something which we never imagined or did before, which was flying”.

3.6 Session 4: Gaming the future

This session was a diegetic prototype exploration in the form of a game jam. The aim of the jam was to suspend the community’s current reality and induce immersion into a new space and world premised on digital gaming, offering inspiration and demonstration of novel possibilities for green energy use. Central to the jam were six games, developed by university students as part of a semester project, each with a theme and story line inspired and based on San culture and folklore. The games were installed on three laptops, with participants playing interchangeably for 1 hour whilst others observed and commented, providing a competitive and lively atmosphere, in addition to a loud sound system playing game soundtracks.

With over 40 community members in attendance, we introduced the games, providing the development background, story-lines and themes of the games, as well as demonstrating the game play and controls. Participants played the games interchangeably, with little interference from the research team bar when needing assistance. There was live commentary and support, with those in attendance demonstrating enthusiasm and excitement as players were navigating through the games. Observers showed great interest, commenting on the gameplay and providing continuous cheering and encouragement. Some players were absorbed in the games, playing for several minutes at a time, whilst others only played once, finding the games either difficult or of little interest, respectively. Whilst playing, players were immersed, attentive and competitive, fully concentrating on the games aiming to complete the games in the fastest possible time. Some players requested to repeat the games, with Kudu Hunter, a hunting game, proving to be popular. After an hour of active play, there was a feedback discussion, during which the participants asserted that the games were “perfect and interesting”. Kudu Hunter was top rated, with appealing graphics and gameplay that is reminiscent of how they used to hunt, with participants stating that playing the game “made us feel backwards in the past”, allowing them to re-establish a temporary, yet meaningful, connection with the past.

4 DISCUSSION AND REFLECTIONS

We discuss and reflect on the above sessions in the following subsections. We highlight relevant and particular considerations from this empirical work, thereby contributing to futuring and temporality discourses from an indigenous community stance and context.

4.1 Duality of futures

Futuring discussions in HCI have primarily been centered on a futuring stance and temporal worldview premised on a singular point of “the present” as evidenced by the widely adopted future cone [26]. However, recent literature has questioned the universality and dominance of such a worldview, with calls to widen the spectrum beyond a singular world and temporal perspective [47],

in addition to looking into the past and opening up “opportunities to a plurality of voices” [46]. We contribute to this argument with our exploration of the future, and its meaning, in Donkerbos, with community members highlighting a dissentient perspective endorsing dual traditional and modern future temporalities. Central to these discussions is the strong relation and association of the past with the future. Indicative of the relevance of the past, and the community’s emotional connection to it, in addition to the desultory of the present. This indicates the projection of a stolen past into the future as one trajectory. This opens up questions for future investigations and consequences for design, namely would those multiple futures be merged or run in parallel and how do we co-design for such dual futures?

4.2 What if ‘What if’ doesn’t work

Tsekleves et al. [50] and Brooms et al. [8] highlight how the *what if* method allows participants to imagine different possibilities and supersede reality and constraints through apprehending and formulating fictional what if scenarios. We employed a similar approach in Donkerbos. However, yielding inverse results, with community members questioning and doubting the appropriateness and logic of the *what if* questions, and showing hesitance towards imagining *what if* scenarios. This illustrates the reluctance and difficulty of initiating imagination through hypothetical scenarios in a local African setting, with preference towards conceivable scenarios. Thus, raising questions and doubt about the *what if* method and its applicability outside its comfort and privileged zone in a rural African indigenous setting. Therefore, it necessitates its appropriation, and the consideration of a wider spectrum of methods. To supplement and compensate the *what if* approach, we undertook a futuring walk. At first, community members were not sure about what to do. However, after clarification and encouragement they showed real enthusiasm and interest, identifying new possibilities. Seemingly, previous interactions and explorations with technologies such as virtual and augmented reality contributed to the community’s openness and thought about unorthodox possibilities. Moreover, walking allowed for the embodiment of new ideas and possibilities, evidenced by the mimicking of a vulture during the demonstration of the flying vulture suit. This indicates, in addition to other studies [4, 29], the potential and effectuality of the method in initiating speculation and creativity.

4.3 Game possibilities

Game jams have been utilised to connect and explore the potential of games in various ways [17]. Similarly, we expand this narrative through the game jam in Donkerbos, with community members playing San cultured games as demonstration of unorthodox possibilities in a rural African context. The games allowed community members to immerse into a world and space premised on digital gaming, demonstrating the affordance of games in aiding the suspension of the “immediate reality” and actualisation of possibilities, in addition to bridging the past and the future. This, fundamentally, illustrates the need for greater, broader and appropriate community exposure to emerging and unorthodox technologies that can transcend communities beyond a one-dimensional reality, invoking further imagination and possibilities.

5 CONCLUSION

In this paper, we narrate futuring explorations and activities with a San community from the Kalahari desert in Namibia. In a series of sessions, divergent speculative design and design fiction methods were employed. During these sessions insight about the applicability and inadequacies of the methods was accumulated. There were varying outcomes, demonstrating mixed effectuality and potential. The future walk and game jam proved successful in their objectives, whereas the *what if* caused doubt and reluctance, necessitating its appropriation and a wider spectrum of futuring methods in an rural African indigenous setting. Additionally, we highlight the community's dissentient dual, past-centric future temporal view, calling for its due consideration in futuring methods and approaches. Particularly, we argue that futuring methods and approaches should take into account different world views, thereby expanding the application of the methods in wider contexts. We see this as an opportunity and point of departure for future work.

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